Starlette
HTTP → Python → HTTP response

Pydantic
Data validation



FastAPI

Why data validation is important

a=10, b=20 → def add(a,b): → 50

10+20=30

return sum

a="ten", b=20 → def add(a,b): → X

"ten" +20 X

return sum

HTTP request → API → Python function

→ a → int

b - int

→

error

API framework → flask

old framework → performance issues
- ① Response time is slow
- ② Application Slow
- ③ Latency issue ↑

(flask) → Lots of unnecessary code → ① more development time

yatra.com/trains

loc1 : Delhi

loc2 : mumbai

date : 5th nov

Submit

→ 2sec

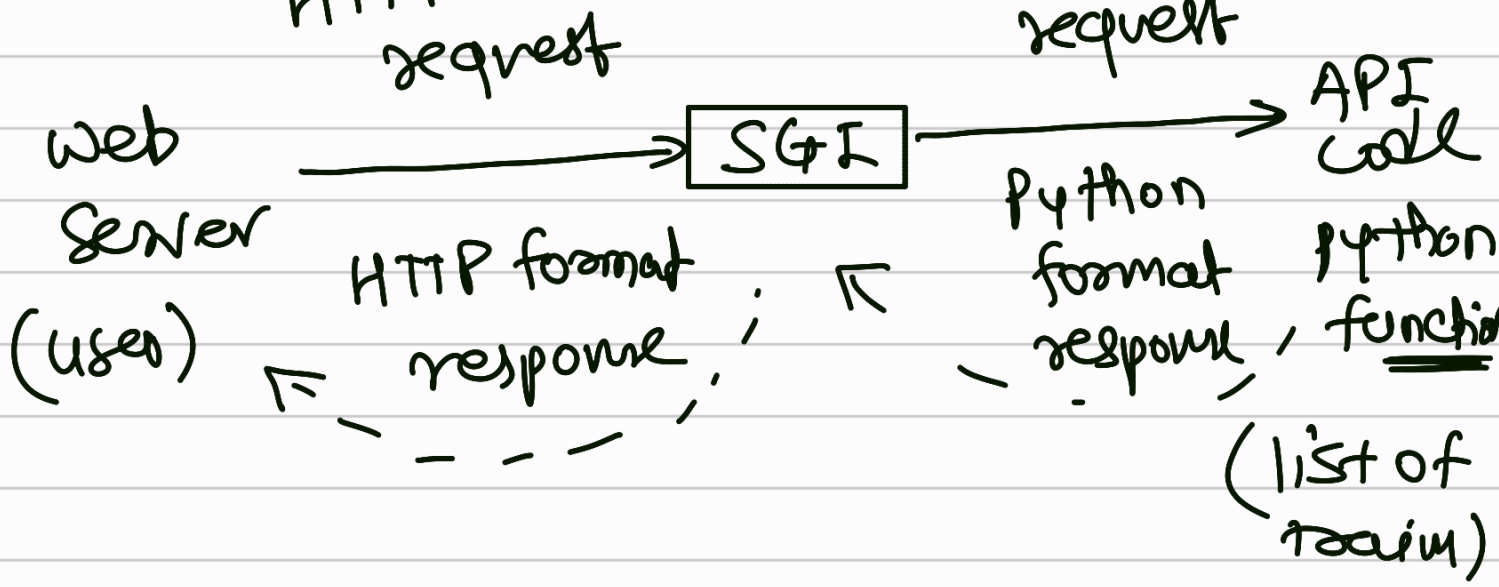
1sec

→ List of available trains

yatra.com/flights

- SGI - Server Gateway interface

HTTP format → Python format



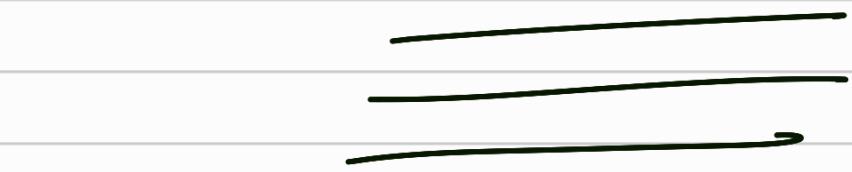
HTTP request format :-

```
Get / pains HTTP/1.1
Host : www.yahoo.com
Content Type : application/json
Content_length : 40
{"loc1" : "Delhi"
 "loc2" : "mumbai"
 "Date" : 5/11/2026 }
```

Python format :

```
resquest.json() —
    { "loc1" : "Delhi"
      "loc2" : "mumbai"
      "Date" : 5/11/2026 }
```

def list_of_trains(loc1, loc2, date):



return list of trains

[ ,

 ,

]

HTTP response format

HTTP/1.1 200 OK

Content-Type: application/json

{ train_name: Golden_temple
time: 6.00 AM

e_time : 11.55 PM

}

Two types of SGI

	WSGI	ASGI
full form	web server gateway interface	Asynchronous Server Gateway Interface
framework	flask	FastAPI
Library	Werkzeug	Starlette
web server	Gunicorn	Uvicorn
Nature	Synchronous	Asynchronous

Blocking Arch.	Non-blocking Arch.

Synchronous — flask (WSGI)

Gunicorn

8.05AM mobile web desktop mobile-ios

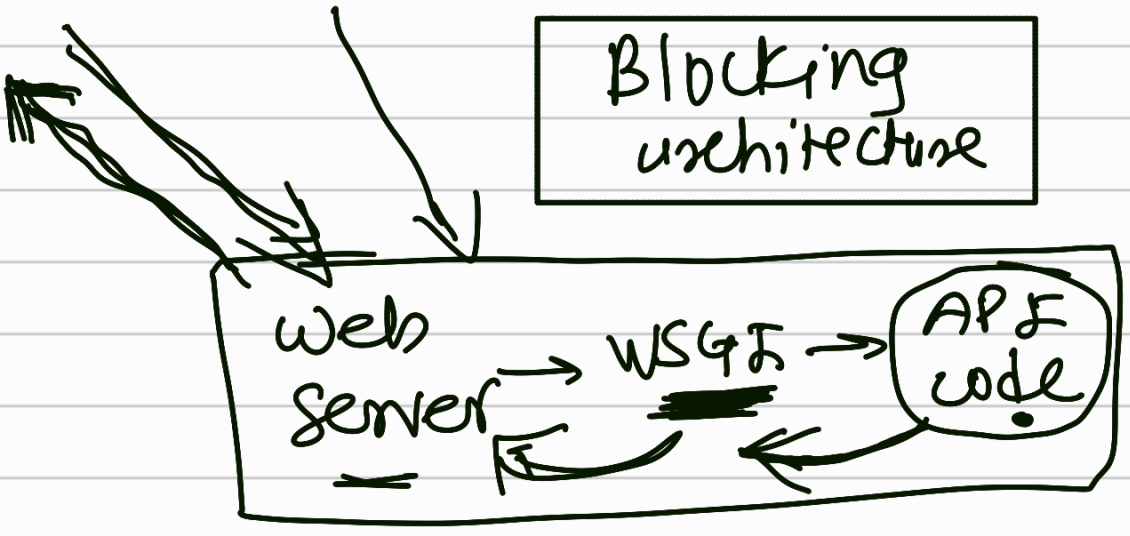
User1 User2 User3 User4

loc1: → wait wait wait

loc2: — " — — " — — " —

Date:

Submit Submit Submit Submit



AWS Application deploy

Running

www.yahoo.com/trains →

127.168.23.5

FastAPI – ASGI → Uvicorn

8.csam mobile-Ar

	web	desk	mob-ios
User 1	User 2	User 3	User 4
loc 1 :	— 11 —	— 11 —	— 11 —
loc 2 :			
date :			

Submit

www.yeabq.com/bcius

web server → ASGI → API code

Advantages of

① Fast to run (ASGI, async)

② Automatic data

FastAPI

Validation
(Pydantic)

③ Auto-generated interactive documentation (Swagger)

④ fast to code

⑤ Seamless integration with libraries Langchain, LangGraph, - -

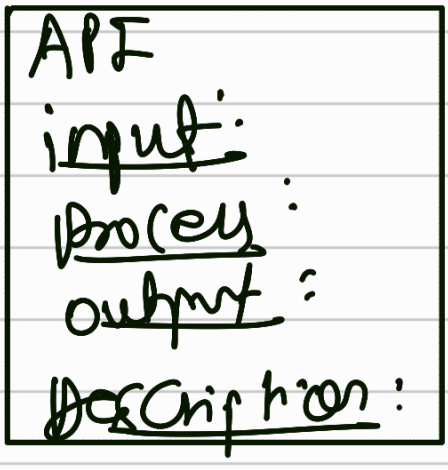
www.yaba.com

/trains

/flights

/bus



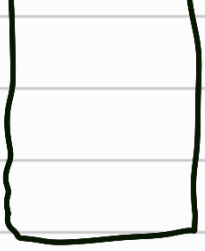


Virtual environment —

1Sdays









1 year

Project 1 :
RAG - Agentic AI
langchain
langgraph
fastAPI
Streamlit

Python 3.11.2

Langchain 1.40

Project 2
Machine learning
numpy
Pandas
scikitlearn
matplotlib
Streamlit
flask

Python 3.14.1

Project 3
Deep learning / NLP
Keras
Pytorch
NLTK
Transformers
Django
fastAPI

Python 3.12.3

Langchain 1.48
1.0.002

Laptop

$ pip install langchain==1.40
$ pip install langchain==1.48

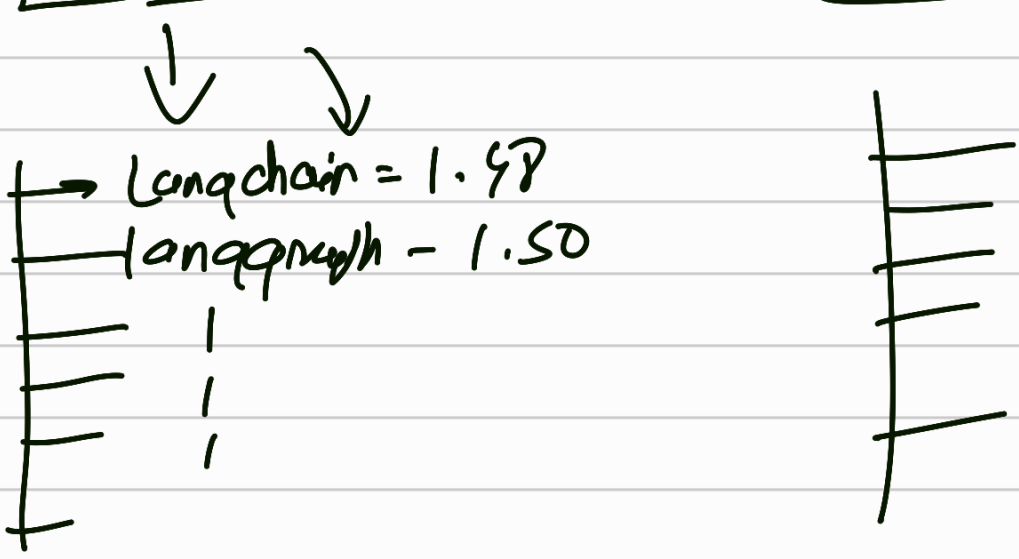
Aug 25 ← 1.0.0.02

Aug 26 ← 1.40

Virtual env

Project 1

Project 2



How to create virtual env.

```
$ python -m venv FastAPI_env        (Linux/mac)
$ source FastAPI_env/bin/activate
$ source FastAPI_env/scripts/activate   (window)
$ pip install fastapi
```

Command to run fastAPI code

```
$ uvicorn file_name:obj_name --reload
```